

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/BEE_4750/hw5-natalie-evelyn5`
```

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
#Problem 1.1 - overall recycling and ash fractions

# % of total mass for each component
mass = [15, 40, 5, 3, 2, 5, 18, 4, 2, 2, 1, 3]

# Combustion ash (%) for each component
ash = [8, 7, 5, 10, 15, 2, 2, 100, 100, 100, 100, 70]

# MRF recycling rate (%) for each component
recycle = [0, 55, 15, 10, 0, 30, 40, 60, 75, 80, 50, 0]

# Overall recycling fraction (R) and ash fraction (A)
R = sum(mass .* recycle) / 10000      # 37.75% -> 0.3775
A = sum(mass .* ash) / 10000          # 16.41% -> 0.1641
```

```
println("Overall recycling fraction R = ", round(R; digits=4))
println("Overall ash fraction A      = ", round(A; digits=4))

# If you also want the recycled and ash masses for each city:
city_total = [100.0, 90.0, 120.0]    # Mg/day for cities 1, 2, 3

recycled_mass = city_total .* R
ash_mass      = city_total .* A

println("\nRecycled mass by city (Mg/day): ", round.(recycled_mass; digits=2))
println("Ash mass by city (Mg/day):      ", round.(ash_mass; digits=2))
```

```
Overall recycling fraction R = 0.3775
Overall ash fraction A      = 0.1641
```

```
Recycled mass by city (Mg/day): [37.75, 33.98, 45.3]
Ash mass by city (Mg/day):      [16.41, 14.77, 19.69]
```

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Here are the following decision variables for the municipal solid waste (MSW) management optimization model.

Variable	Meaning
$W_{ij} \geq 0$	Amount of waste (Mg/day) transported from city i to disposal facility j , where $j \in \{\text{LF}, \text{MRF}, \text{WTE}\}$. These variables represent the primary flows of waste from the cities to each facility.
$R_{kj} \geq 0$	Amount of residual waste or ash (Mg/day) transported from disposal facility k to facility j . This includes non-recycled residue from the MRF sent to LF or WTE, and ash from the WTE sent to LF. These variables represent secondary flows after processing.

Variable	Meaning
$Y_j \in \{0, 1\}$	Operational status of disposal facility j . $Y_j = 1$ if facility j is open (fixed costs and capacity apply), and $Y_j = 0$ if the facility is closed and cannot receive waste. Facilities include LF, MRF, and WTE.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

The objective is to minimize the total daily cost of the MSW system, including: 1. Transportation costs

2. Fixed operating costs for each facility

3. Variable tipping and processing costs

Variables

- $I = \{1, 2, 3\}$: set of cities
- $J = \{\text{LF}, \text{MRF}, \text{WTE}\}$: set of facilities
- $a = 1.5$: transportation cost (\$/Mg · km)
- l_{ij} : distance from city i to facility j
- l_{kj} : distance from facility k to facility j
- c_j : fixed daily operating cost of facility j
- b_j : variable cost per Mg at facility j

From the tables: - $c_{\text{LF}} = 2000$, $c_{\text{MRF}} = 1500$, $c_{\text{WTE}} = 2500$

- $b_{\text{LF}} = 50$, $b_{\text{WTE}} = 60$

- MRF effective variable cost:

$b_{\text{MRF}} = 7 + 40R$ with $R = 0.3775$

The total mass handled at facility j is

$$H_j = \sum_{i \in I} W_{ij} + \sum_{k \in J} R_{kj}.$$

Objective Function

The total transportation cost is

$$C_{\text{transportation}} = a \left(\sum_{i \in I} \sum_{j \in J} l_{ij} W_{ij} + \sum_{k \in J} \sum_{j \in J} l_{kj} R_{kj} \right).$$

The total fixed facility cost is

$$C_{\text{fixed}} = \sum_{j \in J} c_j Y_j.$$

The total variable processing cost is

$$C_{\text{var}} = \sum_{j \in J} b_j H_j.$$

Putting everything together, the complete objective is

$$\min_{W_{ij}, R_{kj}, Y_j} \left[a \left(\sum_{i \in I} \sum_{j \in J} l_{ij} W_{ij} + \sum_{k \in J} \sum_{j \in J} l_{kj} R_{kj} \right) + \sum_{j \in J} c_j Y_j + \sum_{j \in J} b_j H_j \right].$$

This matches the structure of the lecture formulation: transport costs, plus fixed facility costs, plus variable handling costs for each facility.

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Decision variables:

- W_{ij} : waste sent from city i to facility j (Mg/day)
- R_{kj} : residual or ash sent from facility k to facility j (Mg/day)
- Y_j : binary variable indicating if facility j is open (1) or closed (0)

with cities $i \in I = \{1, 2, 3\}$ and facilities $j, k \in J = \{\text{LF}, \text{MRF}, \text{WTE}\}$.

From Problem 1.1 we treat as **parameters**:

- overall recycling fraction $R = 0.3775$
- overall ash fraction $A = 0.1641$

Facility capacities (Mg/day):

- $K_{\text{LF}} = 200$
- $K_{\text{MRF}} = 350$
- $K_{\text{WTE}} = 210$

City waste generation (Mg/day):

- $S_1 = 100, S_2 = 90, S_3 = 120.$

1. City mass-balance constraints

All waste generated in each city must be sent to one or more facilities:

$$\sum_{j \in J} W_{ij} = S_i, \quad \forall i \in I.$$

These equations ensure that no waste is left unassigned or “created” at a city.

2. Facility capacity constraints

Total incoming material to each facility (from cities and other facilities) cannot exceed its capacity. We also link capacity to the binary variable Y_j so that if a facility is closed ($Y_j = 0$) it cannot receive any waste:

$$\sum_{i \in I} W_{ij} + \sum_{k \in J} R_{kj} \leq K_j Y_j, \quad \forall j \in J.$$

Here K_j is the capacity of facility j .

If $Y_j = 0$ the right-hand side is zero, forcing all flows to that facility to be zero. If there is any positive flow, it will be optimal to choose $Y_j = 1$ and incur the fixed cost in the objective.

3. MRF mass balance (recycling and residue)

Let the total mass entering the MRF be

$$H_{\text{MRF}} = \sum_{i \in I} W_{i,\text{MRF}}.$$

A fraction R of this flow is recycled and leaves the system, while the remaining fraction $(1 - R)$ becomes residue. All residue must go either to LF or to WTE:

$$R_{\text{MRF,LF}} + R_{\text{MRF,WTE}} = (1 - R) H_{\text{MRF}} = (1 - R) \sum_{i \in I} W_{i,\text{MRF}}.$$

This constraint enforces conservation of mass at the MRF: all non-recycled waste is accounted for in the residual flows.

4. WTE mass balance and ash generation

Let the total mass entering the WTE be

$$H_{\text{WTE}} = \sum_{i \in I} W_{i,\text{WTE}} + R_{\text{MRF,WTE}}.$$

A fraction A of this incoming mass becomes bottom ash that must be sent to the landfill. We represent this with the residual variable from WTE to LF:

$$R_{\text{WTE,LF}} = A H_{\text{WTE}} = A \left(\sum_{i \in I} W_{i,\text{WTE}} + R_{\text{MRF,WTE}} \right).$$

This constraint ensures that the ash produced by the WTE is consistent with the ash fraction computed in Problem 1.1 and that all ash is routed to the landfill.

5. Domain constraints

All flow variables must be nonnegative:

$$W_{ij} \geq 0, \quad \forall i \in I, j \in J,$$

$$R_{kj} \geq 0, \quad \forall k \in J, j \in J.$$

Facility status variables are binary:

$$Y_j \in \{0, 1\}, \quad \forall j \in J.$$

In practice, we only define R_{kj} for physically meaningful connections (e.g., MRF $\rightarrow$ WTE or LF, WTE $\rightarrow$ LF); all other R_{kj} are either omitted from the model or constrained to zero.

Together, these constraints enforce (i) conservation of mass at cities and facilities, (ii) consistency of recycling and ash generation with the fractions computed in Problem 1.1, and (iii) capacity and operational limits for each disposal option.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
cities = 1:3
facilities = [:LF, :MRF, :WTE]

# waste generated from cities (Mg/day)
waste = Dict{1 => 100.0, 2 => 90.0, 3 => 120.0}

# facility capacities (Mg/day)
K = Dict{:LF => 200.0, :MRF => 350.0, :WTE => 210.0}

# fixed costs ($/day)
C_fixed = Dict{:LF => 2000.0, :MRF => 1500.0, :WTE => 2500.0}

# tipping costs ($/Mg)
C_tipping = Dict{:LF => 50.0, :MRF => 7.0, :WTE => 60.0}

# MRF recycling cost ($/Mg recycled)
C_recycle = 40.0
```

```

# transportation cost ($/Mg-km)
C_transport = 1.5

# distances from city to facility
dist_city_facilities = Dict( (1,:LF) => 5.0, (1,:MRF) => 30.0, (1,:WTE) => 15.0,
                             (2,:LF) => 15.0, (2,:MRF) => 25.0, (2,:WTE) => 10.0,
                             (3,:LF) => 13.0, (3,:MRF) => 45.0, (3,:WTE) => 20.0 )

# distances between facilities
dist_facilities = Dict( (:MRF,:LF) => 32.0, (:MRF,:WTE) => 15.0,
                        (:WTE,:LF) => 18.0 )

# Recycling and ash fractions
R = 0.3775
A = 0.1641

# jump model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Decision variables
# waste from city i to facility j
@variable(model, W[cities, facilities] >= 0)

# MRF residuals to facility j (LF, WTE)
R_mrf = Dict(
    :LF => @variable(model, lower_bound=0.0),
    :WTE => @variable(model, lower_bound=0.0)
)

# ash from WTE to LF
@variable(model, R_to_lf >= 0)

# Y_j binary
@variable(model, Y[f in facilities], Bin)

# constraints

# 1) City mass balance
for i in cities
    @constraint(model, sum(W[i,j] for j in facilities) == waste[i])

```

```

end

# 2) MRF mass balance (residuals)
@constraint(model, R_mrf[:LF] + R_mrf[:WTE] == (1.0 - R) * sum(W[i,:MRF] for i in cities))

# 3) WTE ash generation
@constraint(model, R_to_lf == A * ( sum(W[i,:WTE] for i in cities) + R_mrf[:WTE] ) )

# 4) Facility capacities
# LF from city and residuals
@constraint(model, sum(W[i,:LF] for i in cities) + R_mrf[:LF] + R_to_lf <= K[:LF] * Y[:LF])

# MRF from city
@constraint(model, sum(W[i,:MRF] for i in cities) <= K[:MRF] * Y[:MRF])

# WTE from city and MRF residuals
@constraint(model, sum(W[i,:WTE] for i in cities) + R_mrf[:WTE] <= K[:WTE] * Y[:WTE])

# objective function (total daily cost)

# fixed costs if facility open
fixed = sum(C_fixed[j] * Y[j] for j in facilities)

# transportation cost
transportation = sum(C_transport * dist_city_facilities[(i,j)] * W[i,j] for i in cities, j in facilities)

# transportation for MRF residuals
transportation_MRF = C_transport * ( dist_facilities[:,MRF,:LF] * R_mrf[:LF] + dist_facilities[:,WTE,:LF] * R_to_lf )

# Transportation for ash from WTE -> LF
transportation_ash = C_transport * dist_facilities[:,WTE,:LF] * R_to_lf

# tipping costs on all wastes
tipping_direct = sum( C_tipping[j] * W[i,j] for i in cities, j in facilities )
tipping_residual = C_tipping[:LF] * R_mrf[:LF] + C_tipping[:WTE] * R_mrf[:WTE]
tipping_ash = C_tipping[:LF] * R_to_lf

# MRF recycling processing cost applies to recycled mass: R * H_MRF
recycled_MRF = sum(W[i,:MRF] for i in cities)
mrf_recycling = C_recycle * (R * recycled_MRF)

# total objective

```

```

@Objective(model, Min,
    fixed + transportation + transportation_MRF + transportation_ash + tipping_direct + tipping_indirect
)

# solve & report
optimize!(model)
optimal_value = objective_value(model)

println("Optimal total daily cost = $(round(optimal_value; digits=2)) dollars\n")

println("Facility binary (1= open):")
for f in facilities
    println("  $(f) : $(Int(round(value(Y[f]))))")
end

println("\nCity -> Facility flows (Mg/day):")
for i in cities
    for f in facilities
        println("City $(i) -> $(f) : $(round(value(W[i,f])); digits=2)")
    end
end

println("\nMRF residual flows (Mg/day):")
println("R_mrf_LF  = $(round(value(R_mrf[:,LF])); digits=2)")
println("R_mrf_WTE = $(round(value(R_mrf[:,WTE])); digits=2)")

println("\nWTE ash flows to LF (Mg/day): $(round(value(R_to_lf); digits=2)")

# Facility inflows
inflow_LF  = sum(value(W[i,:LF]) for i in cities) + value(R_mrf[:,LF]) + value(R_to_lf)
inflow_MRF = sum(value(W[i,:MRF]) for i in cities)
inflow_WTE = sum(value(W[i,:WTE]) for i in cities) + value(R_mrf[:,WTE])

println("\nFacility inflows (Mg/day):")
println("LF   : $(round(inflow_LF; digits=2))")
println("MRF  : $(round(inflow_MRF; digits=2))")
println("WTE  : $(round(inflow_WTE; digits=2))")

```

Optimal total daily cost = 27855.48 dollars

Facility binary (1= open):
 LF : 1

MRF : 0
WTE : 1

City -> Facility flows (Mg/day):

City 1 -> LF : 100.0
City 1 -> MRF : 0.0
City 1 -> WTE : 0.0
City 2 -> LF : -0.0
City 2 -> MRF : -0.0
City 2 -> WTE : 90.0
City 3 -> LF : 78.41
City 3 -> MRF : 0.0
City 3 -> WTE : 41.59

MRF residual flows (Mg/day):

R_mrf_LF = 0.0
R_mrf_WTE = 0.0

WTE ash flows to LF (Mg/day): 21.59

Facility inflows (Mg/day):

LF : 200.0
MRF : 0.0
WTE : 131.59

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

According to the optimal solution, MRF will not be used at all, while LF will be used to its full capacity (200 Mg/day), and WTE will be used 131.59 Mg/day (out of capacity 210 Mg/day). This may make sense: despite the relatively inexpensive tipping costs of MRF, there is an added recycling cost of \$40/Mg. Thus, the optimal solution is suggesting to send the waste directly to LF and WTE.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 .

In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

Key information for our two-period stochastic linear program: From our scenario tree (Problem 2.1), we can see that while there is only one possible scenario for the first period, there are four possible scenarios for the second period. Thus, we need to account for these uncertainties.

Decision Variables:

First period:

- $P_{g,1} \geq 0$ = amount of power generated for each generator g in period 1

Second period: * $P_{g,2}^s \geq 0$ = amount of power generated for each generator g in period 2, where scenario $s = 1, 2, 3, 4$

Objective Function:

Our goal is to minimize the cost of generation while meeting the demand AND while considering uncertainty in period 2:

$\min \sum_g C_g P_{g,1} + \sum_{s=1,2,3,4} p_s (\sum_g C_g P_{g,2}^s)$, where: * C_g = variable costs of each generator g (\$/MW) * p_s = probability of scenario s in period 2.

Constraints:

First period:

Generator limits: Each generator cannot produce more than its max capacity, nor less than its min. * $P_{gmin,1} \leq P_{g,1} \leq P_{gmax,1}$

$$\bullet P_{solar,1} \leq CF_{solar,1} * ramp_{solar}$$

Renewable energy sources limits: Solar and wind are reliant/limited by their capacity factors. * $P_{wind,1} \leq CF_{wind,1} * ramp_{wind}$

Demand balance: Demand must be met in period 1. * $\sum_g P_{g,1} = D_1$

Second period: Generator limits: Similar to period 1, generators cannot generate more than max capacity or less than min. * $P_{gmin,2} \leq P_{g,2}^s \leq P_{gmax,2}$

Renewable energy limits: Renewable limits for period 2 depends on the scenarios. * $P_{solar,2}^2 \leq CF_{solar,2}^2 * Cap_{solar} * P_{wind,2}^2 \leq CF_{wind,2}^2 * Cap_{wind}$

Demand balance: Demand must also be met in period 2. * $\sum_g P_{g,2}^2 = D_2^2$

More overall constraints that are applicable to both periods:

ramping constraint: Generators also cannot change output instantaneously, as there are ramping constraints. * $-P_g^{minramp} < P_{g,2}^s - P_{g,1} < P_g^{maxramp}$

nonnegative: Generation cannot be negative. * $P_{g,1}, P_{g,2}^2 \geq 0$

References

List any external references consulted, including classmates.

Problem 1 was discussed with classmate Ally Karanikas—especially the modelling component with JUMP. Professor Srikrishnan's lecture slides on Network Models and Solid Waste Management were consulted for Problem 1, and the slides on Stochastic Optimization were consulted for formulating the stochastic linear model in Problem 2. For JUMP functions and methods used in Problem 1, Professor Srikrishnan's JUMP lab slides + online JUMP documentation were also consulted.